

Asim Ilyas

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PROFESSIONAL SUMMARY

Machine Learning Engineer with a 3.90/4.00 CGPA (97.5%) in Software Engineering from COMSATS University Islamabad and one year of hands-on experience engineering and deploying production AI solutions. Delivered 20+ AI/ML and software projects spanning ensemble learning, explainable AI (XAI), and end-to-end ML pipelines. Ranked 1st (Gold Medal) among 150+ Final Year Projects university-wide. Proficient in Python, Scikit-learn, TensorFlow, PyTorch, FastAPI, Flask, React, and MLOps deployment practices.

CORE COMPETENCIES

Machine Learning · Deep Learning · Ensemble Learning · Explainable AI (XAI) · Model Deployment · Data Preprocessing · Feature Engineering · Model Evaluation · Computer Vision · Natural Language Processing · REST API Development · MLOps · Cross-Validation · Hyperparameter Tuning · Data Visualization · Version Control (Git/GitHub) · Agile Collaboration

EDUCATION

COMSATS University Islamabad

Bachelor of Science in Software Engineering

2022 – 2026

CGPA: 3.90/4.00 (97.5%)

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C#, C, C++, SQL

Machine Learning: NumPy, Pandas, Data Analysis, Feature Engineering, Scikit-learn, TensorFlow/Keras, PyTorch, Classification, Regression, Model Evaluation, Deep Learning

AI Engineering: FastAPI, Flask, REST APIs, Model Deployment, API Integration, Reproducible ML Workflows, Git/GitHub

Computer Vision: OpenCV, Face Detection, Image Preprocessing, Wavelet Features, SVM-based Recognition

Healthcare AI: ECG Preprocessing, 12-Lead ECG Analysis, Explainable AI, Medical AI, Clinical Decision-Support Systems

Tools & Platforms: React, TypeScript, Supabase, Gemini API, REST Integration for AI/ML Dashboards

EXPERIENCE

Junior ML/AI Engineer

TechBitsIT

June 2025 – June 2026

Pakistan

- Completed a 3-month ML/AI internship mastering end-to-end machine learning pipeline fundamentals – data preprocessing, model training, and evaluation – resulting in promotion to Junior ML/AI Engineer.
- Built and deployed ML/AI services using Python, TensorFlow/Keras, Scikit-learn, and FastAPI, integrated into production REST API workflows.
- Collaborated with cross-functional engineering teams across data preprocessing, model training, evaluation, and deployment pipelines.

Machine Learning Instructor

GDGoC (Google Developer Groups on Campus), COMSATS University Islamabad

Feb 2026 – Mar 2026

Pakistan

- Delivered a 1-month Machine Learning bootcamp to 100+ students, covering Python, NumPy, Pandas, and Scikit-learn.
- Mentored students on ML project workflows – preprocessing, training, evaluation, deployment – and on model selection, performance metrics, and debugging.

PROJECTS

Hybrid Multi-Modal Fusion Framework for Explainable Cardiovascular Disease Detection

Python, FastAPI, Scikit-learn, React, TypeScript

- Engineered an AI-powered cardiovascular disease detection system using a lightweight 5-model ensemble, achieving 93% classification accuracy at an approximately 8 MB deployment footprint.

- Architected a complete end-to-end ECG analysis pipeline: Data Collection, Preprocessing, ECG Signal & Image Processing, Feature Extraction, Feature Selection, Multi-Model Training, Hyperparameter Tuning, Ensemble Prediction, Model Evaluation, Post-processing, Automated Report Generation, and Web Deployment.
- Implemented an Explainable AI (XAI) module featuring Random Forest Feature Importance, Lead Contribution Analysis, PQRST Annotation, Classifier Confidence Scores, and Clinical Prediction Explanations for transparent, interpretable results.
- Built secure role-based modules for Admin, Doctor, Technician, and Patient roles, supporting ECG upload, approval workflow, report generation, and patient access.
- Full pipeline visualization: youtu.be/w2KlGjiJKcw Verified: fyp-evaluation.vercel.app

Facial Expression Analyzer

Python, OpenCV, TensorFlow, CNN

- Built a real-time facial emotion detection system using a Convolutional Neural Network (CNN) trained on a labeled facial emotion image dataset (7 emotion classes: happy, sad, angry, surprised, neutral, fear, disgust), achieving 83% classification accuracy.
- Implemented an end-to-end pipeline: Haar Cascade face detection via OpenCV, grayscale preprocessing and resizing to 64×64, CNN-based emotion prediction, and real-time bounding-box visualization on webcam, image, and video inputs.
- GitHub: github.com/asimalyas/Facial_Expression_Analyzer Demo: youtube.com/shorts/e9olDasHyQA

Celebrity Recognition System

Python, OpenCV, Scikit-learn, Flask, Wavelet Transform

- Developed a computer vision classifier across 5 celebrity classes using 2,000+ images, face detection, wavelet feature extraction, and SVM classification, achieving 84% accuracy.
- Integrated the trained model into a Flask web interface for image-based celebrity recognition and inference testing.
- GitHub: github.com/asimalyas/ML-Project Demo: youtube.com/shorts/5PmUVE9-zwM

AI-Integrated Developer Portfolio

React, TypeScript, Vite, Tailwind CSS, Supabase, Gemini API

- Built a responsive portfolio presenting 20+ projects with structured project data, animated sections, and filtering.
- Developed a dynamic admin dashboard backed by Supabase for real-time content management of projects, skills, and updates without redeployment.
- Integrated an AI-powered chatbot using the Gemini API to answer visitor questions about experience and projects in real time.

ACHIEVEMENTS

FYP Final Competition – Gold Medal: Won the Gold Medal at the COMSATS University Islamabad main-campus Final Year Project competition, ranked 1st among 150+ projects campus-wide, for “Hybrid Multimodal Fusion Framework for Explainable Cardiovascular Disease Detection Using ECG Images and Signals.” Verified: fyp-evaluation.vercel.app

Database Project Winner: Ranked 1st after selection from 100+ university projects and a final 12-project competition at COMSATS University Islamabad, 2025.

Machine Learning Project Competition: Earned recognition for Celebrity Recognition System using 5 classes, 2,000+ images, OpenCV, and Scikit-learn.

LEADERSHIP & VOLUNTEER EXPERIENCE

Communication Event Organizer – COMSATS University (Dec 2022): Organized and coordinated the university Communication Event, collaborating with faculty and student volunteers.

Convocation Usher Volunteer – COMSATS University (22nd & 23rd Convocation): Assisted with event coordination and guest management during university convocation ceremonies.

Technical Community Participation – COMSATS University (2024 – Present): Participated in IEEE Student Branch and Robotics Club technical activities; attended GDG DevFest 2024 and GDG DevFest 2026.